Historical Visibility Patterns of European Artists

Ngram K-Means with OpenAlex and Getty cross-source assessment | 1800-2019

1. Objective and datasets

We applied **shape-based K-Means** to historical exact-name visibility trajectories for European painters and asked whether the patterns were also observable in academic-mention and indexed provenance data. The dictionary contains **1,370 Wikidata-linked artists**. Three sources measure different quantities: English **Google Books Ngram** name frequency (20200217 corpus), **OpenAlex** art-topic artist-work mentions, and **Getty** provenance title-transfer or catalog records. They were never added together.

Ngram supplied 73,523 sparse artist-year records (877 artists with exact-name hits). OpenAlex retained 15,679 artist-work matches. Getty attempted all 1,370 artists, accepted 495 identity mappings, and found usable years for 23,007 / 79,715 indexed events.

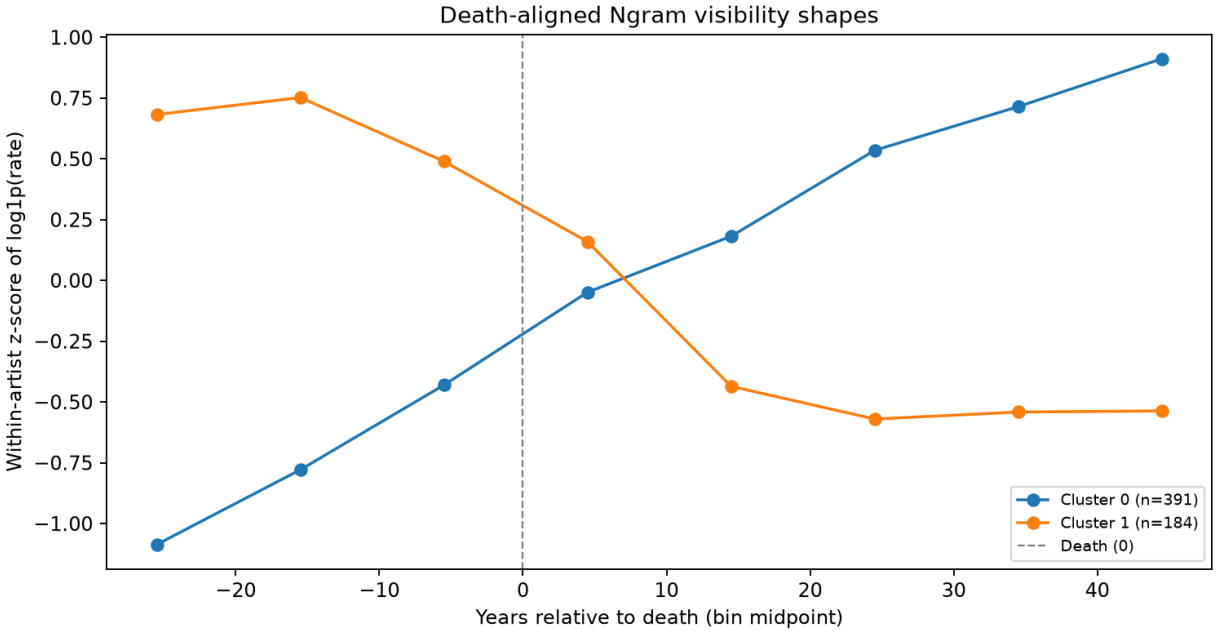
2. Methodology

A1: Compute decade frequency from matching *n*-gram annual corpus denominators; apply log1p and per-person z-standardization; test K=2-6 using silhouette and initialization adjusted Rand index (ARI). Within confirmed complete scans, omitted years are treated as zero exact-name matches only as an explicit model assumption.

A2-A3: Mark suspicious pre-birth hits without deleting artists. Recompute annual data into eight death-relative ten-year bins (death minus 30 to plus 49 years), retaining only fully observed windows with >=4 occupied bins and >=20 hits. Refit calendar and death-aligned K-Means on the *same* eligible artists, with a no-warning sensitivity run.

B-C: Freeze A3 labels and assess OpenAlex normalized document matches and Getty dated title transfers/catalog records on the identical relative-year axis. Account for source-specific coverage and missing dates; no source fusion model is fitted.

Figure 1. Ngram death-aligned cluster-center trajectories

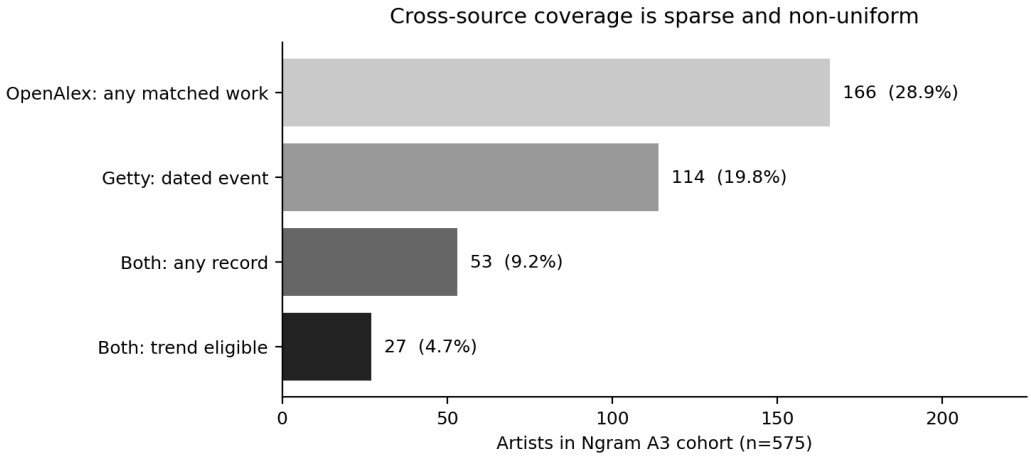


Cluster 0: increasing posthumous relative visibility; Cluster 1: higher pre-death peak then decline. Figure is an unmodified copy of the recorded A3 result.

3. Main results

Analysis	Frozen measured result
A1 calendar Ngram	876 artists; K=2; silhouette 0.3026; initialization ARI 0.9941
A2 quality audit	53 name warnings; birth cohort association Cramer's V = 0.423
A3 death-aligned	575 artists; K=2; silhouette 0.3537; initialization ARI 1.000
A3 membership	Cluster 0: 391; Cluster 1: 184
A3 without warnings	541 artists; K=2; ARI vs original labels 0.9774
OpenAlex assessment	166 any; 91 trend-eligible (>=3 works, >=2 bins)
Getty assessment	291 mapped; 114 dated in-window; 87 trend-eligible
Joint source coverage	27 / 575 (4.7%) trend-eligible in BOTH external sources

Figure 2. External-source coverage in the A3 cohort



OpenAlex trend-eligible: 91 artists; Getty trend-eligible: 87 artists.
OpenAlex and Getty single-source bars are not nested; "Both" denotes the intersection.

4. Interpretation and limitations

The model distinguishes **descriptive name-visibility shapes**, not verified art-historical types. Birth-era effects remain and homonymous names may contaminate counts. OpenAlex matches are sparse and not fully manually disambiguated. Getty lacks usable years for **56,708 / 79,715 (71.1%)** of its indexed records; missing Getty events are **not** zero historical sales. Only **27** artists satisfy both external trend thresholds, so a three-source joint K-Means would be strongly selected and is not recommended. Results are descriptive, not causal.

5. How to run the code

From the **artist_ngram** repository root, in a Python environment with pandas, numpy, matplotlib and scikit-learn, preserve the existing A1-A3/B/C outputs and run:

```
python scripts/ml/build_section5_figures.py --a3-run "results/ml/ngram_death_aligned/run_20261017_134328"
--stage-b-run "results/ml/ngram_openalex_validation/run_20261017_200119" --stage-c-run
"results/ml/ngram_getty_validation/run_20261017_202626_418091"
```

The plotting step reads frozen outputs, checks keys and source hashes, and writes figures, an auditable case-study CSV and a summary into a NEW run folder under results/ml/section5_final/. Full upstream reproduction instructions are in Task5_Teammate_Handoff.md.